

BIALYSTOK, Poland

WMO: 122950

Lat: 53.1073N

Lon: 23.1623E

Elev: 151

StdP: 99.52

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-18.5	-15.1	-20.9	0.6	-18.3	-17.6	0.8	-14.2	8.1	-0.5	7.3	-1.6	1.5	50	0.532

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.3	28.9	20.2	27.1	19.0	25.3	18.1	21.2	26.7	20.2	25.2	19.2	23.7	2.8	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.4	14.4	23.7	18.4	13.5	22.5	17.4	12.7	21.4	62.2	26.8	58.6	25.1	55.1	23.7	24.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.7	5.9	5.2	DB	-21.4	31.8	4.2	1.3	-24.4	32.7	-26.9	33.5	-29.3	34.3	-32.3	35.2	(4)
(5)			WB	-21.6	23.1	4.1	1.1	-24.6	23.9	-27.0	24.5	-29.4	25.2	-32.4	26.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	7.3	-3.5	-2.2	1.0	7.4	12.5	15.7	18.0	17.6	12.3	7.3	2.7	-1.5
	DBStd	8.69	5.78	5.48	4.42	4.15	3.86	3.32	3.07	3.09	3.56	3.92	4.33	5.30
	HDD10.0	1853	419	341	280	98	19	1	0	0	16	102	221	357
	HDD18.3	4103	678	574	538	327	184	94	46	52	182	343	470	615
	CDD10.0	873	0	0	1	21	96	170	247	235	86	17	1	0
	CDD18.3	81	0	0	0	0	3	14	34	28	2	0	0	0
	CDH23.3	892	0	0	0	6	54	171	302	330	29	0	0	0
	CDH26.7	199	0	0	0	0	3	35	78	79	4	0	0	0
Wind	WSAvg	2.5	2.8	2.8	2.8	2.6	2.4	2.3	2.2	1.9	2.1	2.5	2.8	2.8
Precipitation	PrecAvg	609	34	29	34	38	61	68	85	67	53	48	44	40
	PrecMax	934	110	59	82	85	114	148	187	209	167	201	82	88
	PrecMin	444	4	1	6	4	9	22	5	5	13	3	9	11
	PrecStd	104	20	15	17	19	27	30	43	36	30	36	18	17
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.3	9.5	15.4	24.1	26.9	30.0	31.4	31.5	26.7	20.1	13.2	8.9
		MCWB	5.9	6.9	10.1	14.6	17.5	20.6	21.3	20.8	18.2	15.0	11.2	7.7
	2%	DB	5.9	7.7	12.1	20.0	24.9	27.4	28.8	28.8	23.4	16.8	11.2	7.3
		MCWB	4.9	6.0	7.6	12.4	16.6	19.2	20.4	20.2	17.0	13.7	9.6	6.3
	5%	DB	4.6	6.1	9.4	17.4	22.8	25.2	26.6	26.9	20.8	14.7	9.5	6.0
		MCWB	3.7	4.6	6.4	10.8	15.4	17.9	19.5	18.8	15.7	12.2	8.4	5.2
	10%	DB	3.1	4.5	7.3	15.4	20.6	23.0	24.5	25.0	18.5	13.2	8.3	4.6
		MCWB	2.4	3.4	5.1	9.8	14.3	16.8	18.5	17.7	14.5	11.3	7.4	3.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.2	7.6	10.5	15.5	19.1	21.7	23.0	22.7	19.4	15.7	11.7	8.0
		MCDB	6.8	8.8	14.1	22.7	24.9	27.9	28.8	28.5	24.8	19.2	12.8	8.5
	2%	WB	5.0	6.2	8.5	13.4	17.6	20.3	21.3	21.1	17.5	14.0	9.9	6.6
		MCDB	5.8	7.4	11.6	18.7	23.1	25.4	26.3	26.9	21.5	15.9	10.9	7.2
	5%	WB	3.7	4.7	6.8	11.9	16.2	18.8	20.3	19.9	16.3	12.8	8.6	5.2
		MCDB	4.5	5.9	8.8	16.0	21.2	23.2	24.9	25.3	19.9	14.5	9.3	6.0
	10%	WB	2.5	3.3	5.3	10.5	15.0	17.5	19.3	18.7	15.1	11.5	7.5	4.0
		MCDB	3.1	4.4	7.1	13.9	19.3	21.6	23.5	23.3	17.9	13.0	8.2	4.6
Mean Daily Temperature Range	5% DB	MDBR	4.6	5.5	7.6	10.5	11.6	10.9	10.3	11.7	9.4	7.6	4.6	4.3
		MCDBR	4.5	6.2	10.9	14.4	15.1	14.5	13.4	14.9	12.9	10.1	6.3	4.6
		MCWBR	4.1	5.1	7.3	7.8	7.6	7.1	6.0	6.8	7.0	6.7	5.1	4.0
	5% WB	MCDBR	4.3	6.0	10.1	13.0	13.0	12.6	11.7	13.0	11.4	9.3	5.7	4.5
		MCWBR	4.0	5.2	7.2	7.5	6.9	6.7	5.9	6.5	6.5	6.6	4.9	4.1
Clear-Sky Solar Irradiance	taub	0.304	0.335	0.370	0.417	0.404	0.397	0.431	0.427	0.386	0.358	0.346	0.304	
	taud	2.211	2.198	2.223	2.167	2.269	2.312	2.233	2.240	2.323	2.383	2.278	2.249	
	Ebn at Noon	653	750	804	810	846	855	814	793	782	725	598	582	
	Edh at Noon	66	91	110	132	125	122	129	122	99	76	63	54	
All-Sky Solar Radiation	RadAvg	0.67	1.41	2.65	4.07	5.25	5.52	5.17	4.47	3.12	1.67	0.64	0.43	
	RadStd	0.12	0.24	0.39	0.52	0.52	0.47	0.54	0.40	0.46	0.32	0.06	0.07	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	+0.70	N/A	N/A	N/A	N/A
(47) Regional (17 neighbors)		+0.46	N/A	N/A	N/A	N/A	+0.55	N/A	-152	+83	N/A

Nomenclature: See separate page